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Exercise 3C

1) mass and velocity
2) Linear momentum of a body is the product of its mass and velocity.
4)

Let the initial momentum = mu
let the final momentum = mv

So $\Delta p = (mv - mu) = m(v-u)$

so rate of change of momentum $= \frac{\Delta p}{\Delta t} = \frac{m(v-u)}{t}$

but as acceleration $= \frac{v-u}{t}$, then $\frac{\Delta p}{\Delta} = m \times a$

5) (i) 1:1 as their masses are the same
(ii) 1:2 as 2v is double of v.

6) (i) 1:2 as masses are different
(ii) 1:1
(iii) 1:1

8)

11) You can obtain the Newtons first law by Newton's second Law as Newton's ~~first law~~ second law shows you that F = mxa and what Newtons first law shows us is that there is F that causes a motion and Newtons second law helps us calculate that.

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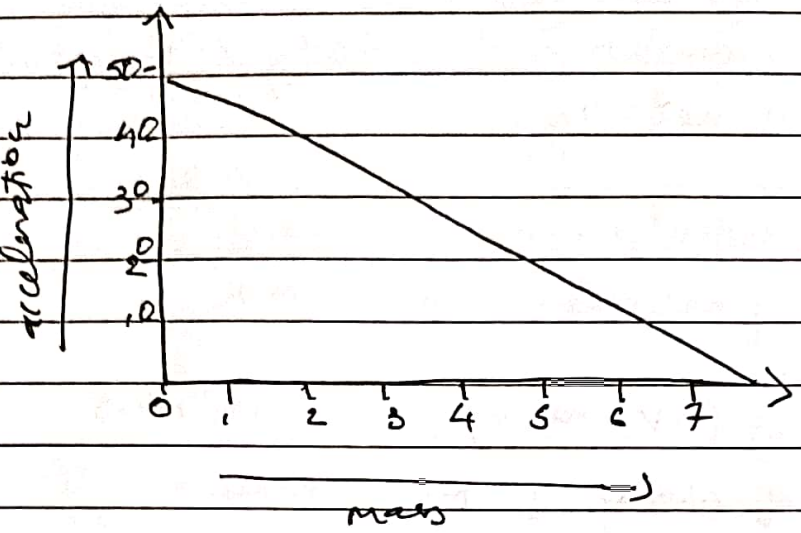
13) So as we know $\boxed{a \propto m}$. As if the mass increases the acceleration decreases.



15) One dyne is the force which when acts on a body of mass 1g, produces an acceleration of $1 cm s^{-2}$, 1 dyne = 1g × 1 cm/s²

16) S.I unit of force = Newton C.G.S unit of force = dyne. They are related because both are units of force and 1 Newton = 10^6 dyne.

18) (a) So we can see that

Force × time = mass × (change in velocity)

so if the cricketer pulls his hand back, he is increasing the ~~time~~ time taken to stop the ball which will decrease the force that will be exerted on his hand.

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(b) So they landing on sand, the athelete is ~~throw~~ increasing the time taken to come to stop as it will reduce the impact force.

Numericals :

2) $\boxed{50g = 0.05kg}$

$0.5 = 0.005 \times u$

$u = \frac{0.5}{0.005}$

$= 10$

$\therefore u = 10 \, m/s$

3) $15 = 2 \times u$

$u = \frac{15}{2}$

$= 7.5 \, m/s^2$

6 (a) $F \times t = m \times (v-u)$

(i) $10 \times 3 = 2 \times v - 0$

$30 = 2 \times v$

$v = \frac{30}{2}$

$v = 15 m/s$

(ii)

$\Delta p = m(v-u)$

$= 2 \times 15$

$= 30 \, kgm/s$

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7)

(i) $F.t = m \times (v - u)$

$v = \frac{d}{t}$

so $v = \frac{100}{5} = 20 \, m/s$

(iii)

$F.10 = 100 \times 20$

$F \times 10 = 2000$

$F = \frac{2000}{10} = 200 \, N$

(ii)

$\boxed{F = m \times a}$

so

$200 = 100 \times a$

$a = 2 \, m/s^2$

8) ~~as acceleration = slope~~

~~$a = \frac{v}{t}$~~ ~~$\sqrt{400 \times 25}$~~

~~$= 100000$~~

~~slope $= \sqrt{(v^2 + t^2)}$~~

~~$= \sqrt{400 + 25}$~~

~~$= \sqrt{425}$~~

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8) ~~Slope~~ = Acceleration = $\frac{v}{t}$

$= \frac{20}{5}$

$= 4\ m/s^2$

So

$F = 0.1\ kg \times 4\ m/s^2$

$= 0.4\ N$

$\therefore F = 0.4N$

$0.15 \times 25 = 75$

10 a) $F \times t = m \times (v-u)$

So

$F \times 0.03 = 0.15 \times -25$

~~F × 0.03 =~~

$F = \frac{0.15 \times -25}{0.03}$

$F = \frac{15 \times 25}{3}$

$= -125N$

11) $F \times 0.1 = 2 \times 2$

$F = \frac{4}{0.1}$

$= 40\ N$

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